INCH-POUND

MIL-PRF-1/985C
10 May 2007
SUPERSEDING
MIL-E-1/985B
15 July 1977

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5U4GB

This specification sheet is inactive for new design after 21 July 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this specification sheet and MIL-PRF-1.

DESCRIPTION: Rectifier, full wave, high vacuum.
Outline - - - 12-16.
Base - - - B5-121.
Envelope - - - T12.
Cathode - - - Coated filament.

Base connections:

Pin no.	- - -	1	2	3	4	5	6	7	8
Element	- - -	nc	f	---	2a	---	1a	---	f

ABSOLUTE RATINGS:

Parameter:	Ef	Epp/a	epx	ib	Io	R1	CL	i(surge)	Alt
Unit:	V ac	V ac	v	ma	mA dc	Ohms	μF	a	ft
Maximum:	5.5	---	1,700	1,100	300	---	---	5.0	10,000
Minimum:	4.5	---	---	---	---	---	---	---	---
Test conditions:	5.0	550	---	---	---	1,950	10	---	---

GENERAL:

Qualification - Not required.

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TABLE I. Requirements and tests.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 7)	Symbol	Limits		Unit
					Min	Max	
	First article test						
1031	Low-frequency vibration	No voltages	---	---	---	---	---
	Conformance inspection, part 1						
1353	Operation of rectifiers (1)	See notes 2 and 3	0.65	Io	285	---	mA dc
1231	Emission	Eb = 75 V dc See note 4	0.65	Is	350	---	mA dc
	Conformance inspection, part 2						
---	Alignment	See note 5	---	---	---	---	---
1301	Filament current		---	If	2.7	3.3	A
1353	Operation of rectifiers (2)	Ef = 4.5 V ac	---	$\frac{\Delta Io}{If}$	---	10	%
1211	Insulation of electrodes		---	---	---	---	---
1101	Secureness of base, cap, or insert		---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---
	Conformance inspection, part 3						
---	Life-test provisions	Group B, see note 6	---	---	---	---	---
---	Life-test end point (500 hours)						
1353	Operation of rectifiers (1)		---	Io	275	---	mA dc

NOTES:

1. Tube shall be operated in a vertical position. Tube may be mounted in a horizontal position provided pins no. 2 and 4 are in a vertical plane.
2. In a full-wave circuit Zp/a shall be adjusted such that a tube having Etd = 44 V dc at 225 mA dc per anode gives Io = 300 mA dc.
3. This test shall be performed at the conclusion of the holding period.
4. Test each section separately. Anode voltage = 0 for section not under test.
5. The plane of the filament shall be parallel to the plane passing through the centerline of pins No. 2 and 4 within ±22.5°.
6. The values of R1 and CL given in test conditions may be considered as approximate and shall be adjusted initially to give Io = 300 mA dc with ib ≥1.1A.
7. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-PRF-1, this document references MIL-STD-1311.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, CR4, MI
Navy - AS, CG, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5960-2007-022)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST online database at: http://assist.daps.dla.mil.

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